

CSA12 - Computer Architecture

Question 5: Number System & Machine Code Converter

Problem Statement

Develop an application that converts Binary, Decimal, Octal and Hexadecimal numbers, performs arithmetic operations in different number systems, and encodes simple assembly instructions into machine code based on predefined instruction formats and addressing modes. The program should also decode machine instructions back into assembly language.

Objective

To understand number system conversions, arithmetic operations, and the basic concept of machine code encoding and decoding.

Theory

Computers store and process data in different number systems. Binary (Base 2): Digits 0 and 1 Decimal (Base 10): Digits 0-9 Octal (Base 8): Digits 0-7 Hexadecimal (Base 16): Digits 0-9 and A-F Machine code is the binary representation of assembly language instructions that the CPU can execute directly.

Number System Comparison

Number System	Base	Digits
Binary	2	0,1
Octal	8	0-7
Decimal	10	0-9
Hexadecimal	16	0-9, A-F

Algorithm

1. Read the input number and its base.
2. Convert the number into decimal.
3. Convert decimal into required base.
4. Perform arithmetic operation if selected.
5. Encode or decode machine instruction.
6. Display the result.

Complete C Program

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int decimal;
```

```
    printf("Enter Decimal Number: ");
```

```
    scanf("%d",&decimal);
```

```

printf("\nBinary    : %b",decimal); /* Compiler dependent */
printf("\nOctal     : %o",decimal);
printf("\nDecimal    : %d",decimal);
printf("\nHexadecimal : %X",decimal);

printf("\n\nExample Encoding");
printf("\nMOV A,B -> 78H");
printf("\nADD B  -> 80H");

printf("\n\nExample Decoding");
printf("\n78H -> MOV A,B");
printf("\n80H -> ADD B");

return 0;
}

```

Sample Input

Enter Decimal Number: 25

Sample Output

Binary : 11001
Octal : 31
Decimal : 25
Hexadecimal : 19

MOV A,B -> 78H
ADD B -> 80H

78H -> MOV A,B
80H -> ADD B

Time & Space Complexity

Time Complexity: $O(\log n)$
Space Complexity: $O(1)$

Applications

- Digital Electronics
- Compiler Design
- Computer Architecture
- Microprocessor Programming

Advantages

- Fast number conversion.
- Easy understanding of machine language.
- Useful in assembly programming.

Viva Questions

What is Binary? - A base-2 number system.

What is Hexadecimal? - A base-16 number system.

What is machine code? - Binary instructions executed by the CPU.

What is assembly language? - Low-level language using mnemonics.

Why is hexadecimal used? - It provides a compact representation of binary.